

# Shobhit Upadhyaya

Principal AI Scientist — LLM, RAG & Conversational Systems  
Bangalore, India

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## Summary

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Principal AI Scientist with 16+ years (9+ in ML/AI) specializing in production-scale LLM and RAG systems. Proven expertise in designing conversational AI architectures with measurable business impact, improving retrieval quality, reasoning, and automation at scale. Built enterprise AI systems serving 148K users over a multi-million document corpus, increasing deflection from 26% to 55%.

## Professional Experience

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### Oracle — Principal Data Scientist

Aug 2021 – Present

- Designed end-to-end RAG architecture (retrieval, ranking, prompt orchestration, evaluation) for a production conversational AI system serving 148K users and 24K queries/day over a multi-million document corpus
- Led development of multi-turn conversational AI system, improving query deflection from 26% to 55%
- Built hybrid reasoning architecture combining structured dialogue flows with LLM-based reasoning for ambiguity resolution (+11% deflection)
- Recovered ~10% failed queries via query simplification and multi-query fallback, improving deflection by ~6%
- Established LLM evaluation frameworks to measure response quality, retrieval accuracy, and user satisfaction
- Optimized retrieval and generation latency for real-time performance in large-scale conversational systems
- Improved response quality through structured LLM output, increasing self-completion rates by +2–3%
- Conducted A/B experimentation across query understanding, prompt strategies, and response formats to optimize retrieval quality, response accuracy, and user experience

### Societe Generale — Lead Data Scientist

May 2018 – Aug 2021

- Built ML-based cash reconciliation system using multi-class classification (93% F1-score)
- Automated financial categorization workflows, reducing manual effort by ~7 FTE
- Improved operational efficiency and accuracy in large-scale financial data pipelines

### Tecnotree — System Specialist (ML)

Dec 2014 – May 2017

- Developed churn prediction model (89% F1-score) for telecom customers
- Built customer segmentation models (K-Means) using behavioral and usage data

### Earlier Experience

ITC Infotech (2012–2014) — Sasken (2010–2012) — Nanolets (2009–2010)

## Technical Skills

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**AI/ML:** LLMs, RAG, NLP, Conversational AI, Query Understanding, Multi-turn Dialogue Systems

**Systems:** Prompt Engineering, LLM Evaluation Frameworks (RAGAS), A/B Experimentation, Model Monitoring

**Infrastructure:** Vector Search (OpenSearch / FAISS), Scalable AI Systems, Latency Optimization

**Programming:** Python (NumPy, Pandas, Scikit-learn, Pytorch)

## Achievements

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- USAID Forecasting Challenge (2020) — 3rd Rank
- Ericsson ML Challenge (2019) — 2nd / 4120 participants
- MakeMyTrip Data Science Challenge (2018) — 3rd / 3556 participants
- BrainWaves ML Hackathon (2018) — Finalist

## Education

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Master of Computer Applications (MCA), UPTU

2005 – 2008